

Correct Answer:

```
treat:time
```

Explanation

In a standard **Difference-in-Differences (DiD)** regression model:

```
R
lm(outcome ~ treat * time, data = df)
```

The coefficient on the **interaction term** `treat:time` represents the **Average Treatment Effect on the Treated (ATT)**.

Interpretation of Each Coefficient

Coefficient	Interpretation
(Intercept)	Baseline outcome for the control group in the pre-period
treat	Pre-treatment difference between treated and control groups
time	Change over time in the control group (common time trend)
treat:time	ATT — the <i>additional</i> change in the treated group after treatment (the causal effect of interest)

Why `treat:time` is the ATT

The DiD estimator isolates the causal effect by subtracting the control group’s time change from the treatment group’s time change:

$$ATT = (Treated_{post} - Treated_{pre}) - (Control_{post} - Control_{pre})$$

This is exactly what the `treat:time` coefficient captures.

Quick Code Reference

R

```
did_model <- lm(outcome ~ treat * time, data = df)
summary(did_model)
```

Look for the row labeled `treat:time` — that is your DiD estimate (ATT).

Would you like help interpreting a full regression output, adding covariates, or creating a visualization of the DiD effect?